

Observability (Scenarios)

Key Metrics

Metric	Type	Description
jvm_cpu_recent_utilization_ratio	Gauge	Recent CPU usage by JVM
jvm_thread_count	Gauge	Number of active JVM threads
jvm_memory_used_bytes	Gauge	Memory currently used by the JVM
jvm_gc_collection_seconds_total	Counter	Total time spent in garbage collection
http_requests_total	Counter	Total number of HTTP requests received
http_server_request_duration_seconds_bucket	Histogram	Duration of HTTP server requests
http_client_request_duration_seconds_bucket	Histogram	Duration of outgoing HTTP client requests
http_errors_total	Counter	Total number of HTTP errors (4xx, 5xx)
process_resident_memory_bytes	Gauge	Resident memory used by the process
node_memory_MemAvailable_bytes	Gauge	Available memory on the node

Scenarios

● Low Load

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=100&faults=0' | vegeta attack -duration=60s --rate=10 | vegeta report
```

- **Metrics to observe:**

- `jvm_cpu_recent_utilization_ratio` : low CPU usage (max ~0.0175)
- `jvm_thread_count` : low thread count
- `http_server_request_duration_seconds_bucket` : stable, low latency (~3.25 ms peak)
- `http_client_request_duration_seconds_bucket` : stable, low latency (2-6 ms)

● Mid Load

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=100000&faults=0' | vegeta attack -duration=60s --rate=20 -timeout=120s | vegeta report
```

- **Metrics to observe:**

- `jvm_cpu_recent_utilization_ratio` : higher CPU usage (peak ~0.118)
- `jvm_thread_count` : increased threads, mostly waiting/runnable
- `http_server_request_duration_seconds_bucket` : gateway and math-service show slowdowns, some 503 errors
- `http_client_request_duration_seconds_bucket` : increased latency, some request timeouts

● High Load

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=1000000&faults=0' | vegeta attack -duration=180s --
```

```
rate=60 -timeout=120s | vegeta report
```

- **Metrics to observe:**

- `jvm_cpu_recent_utilization_ratio` : high CPU usage
- `jvm_thread_count` : high, thread contention visible
- `http_server_request_duration_seconds_bucket` : almost all requests fail or timeout
- `http_client_request_duration_seconds_bucket` : extreme latency, 0% success

⚠ **Mid Load with Faults (40%)**

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=1000000&faults=40' | vegeta attack -duration=180s --  
rate=3 -timeout=120s | vegeta report
```

- **Metrics to observe:**

- `jvm_cpu_recent_utilization_ratio` : moderate to high CPU usage
- `jvm_thread_count` : elevated threads
- `http_server_request_duration_seconds_bucket` : mixed success/failures (200, 500, 503)
- `http_client_request_duration_seconds_bucket` : most requests fail or timeout